

NOVACAP FINANCIAL TECHNOLOGIES

Principal Strategist — 90-Day FinTech Arbitrage Academy

Principal Strategist — 90-Day FinTech Arbitrage Training Simulator

"This is not a course. This is a simulation of the most demanding trading desk in the region."

A hard-mode, self-contained algorithmic trading training simulator. You write Python code against a live mock exchange, log trades to a double-entry SQLite ledger, pass AST static audits, and face ruthless EOD evaluations. Score 100 or you don't advance.

 What You Get

One file. Zero dependencies. 90 days of rigorous training.

SPEC	DETAIL
File size	~4,000 lines of Python
Dependencies	None — standard library only
Python	3.8+ (any OS: Windows, macOS, Linux)
Setup	<code>python arbitrage_academy.py</code> — that's it
Interface	CLI (cmd.Cmd) + full Web Dashboard on <code>:8081</code>
Mock Exchange	HTTP server on <code>:8080</code>

 What You'll Master

Phase 1 — Days 1–30: Data Ingestion & Risk Math

- Fetch order books from REST APIs with error handling
- Calculate SMA, EMA, Bollinger Bands, RSI
- Log all data to a double-entry SQLite ledger
- ASTM static code audits on every submission

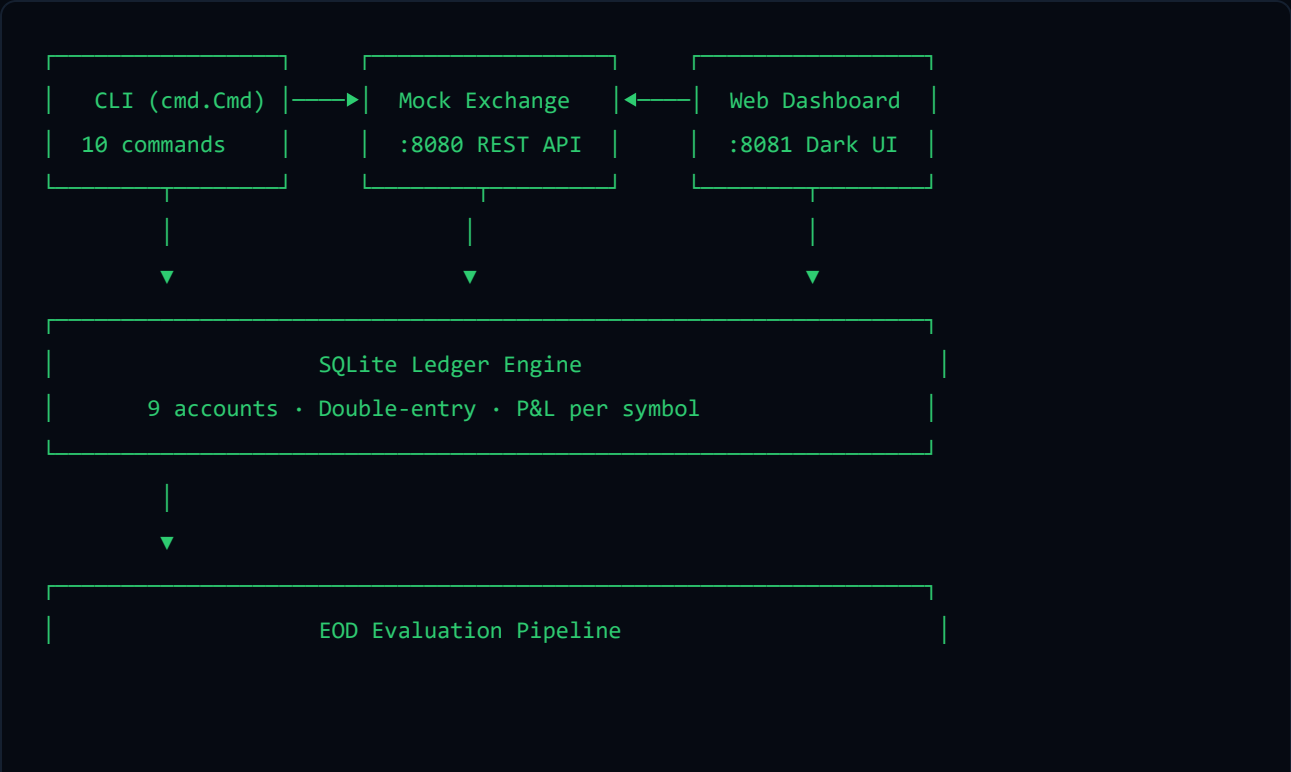
Phase 2 — Days 31–60: Arbitrage Logic

- Cross-exchange spread detection (PRIMARY vs SECONDARY)
- Real-time triangular arbitrage (BTC → ETH → USDT → BTC)
- Position sizing and Kelly Criterion
- Latency simulation and slippage models

Phase 3 — Days 61–90: Slippage & SOCPA Compliance

- SOCPA-compliant audit trails and reconciliation
- IFRS 9 financial instrument reporting
- VaR, CVaR, and stress testing
- Full compliance report generation

Architecture Overview



AST Audit → Subprocess Tests → Ledger Verify → Score (0-100)

11 Interconnected Modules

#	MODULE	PURPOSE
1	Color/ANSI	Terminal output formatting with 16-color palette
2	Principal Strategist	Clinical-tone persona with observations & consequences
3	MockExchangeServer	Threaded HTTP server simulating 5 symbols
4	MockPriceGenerator	Realistic bid/ask spread generation
5	LedgerEngine	Double-entry SQLite with 9 accounts
6	CodeAuditor	<code>ast</code> -based static analysis (9 check categories)
7	StateManager	JSON persistence for 90-day progression
8	WorkspaceGenerator	Per-day file generation (briefing + boilerplate + tests)
9	EvaluationEngine	Subprocess test runner + scoring
10	ArbitrageAcademyCLI	<code>cmd.Cmd</code> with 10 commands
11	WebUIServer	Threaded HTTP dashboard with live polling

 Screenshots

Web Dashboard — Live Market Feed & Controls

NovaCap Principal Strategist

● Day 1Phase 1

BTC/USD \$67,306.92ETH/USDT \$3,146.82BTC/USDT \$68,188.97

Current Day

1 / 90

Phase

Data Ingest

Days Passed

0

Cum. Score

0.0%

```
| ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐ |
| [▶ Start Day] [⊖ EOD] [▶ Run] [→ Advance] |
| └──────────────────────────────────────────────────────────┘ |
```

Mock Exchange REST API — Live Order Book

```
GET /v1/ticker?symbol=BTC/USD
{
  "symbol": "BTC/USD",
  "price": 67306.92,
  "change_24h_pct": -4.07,
  "volume_24h": 5739.1,
  "timestamp": "2026-06-18T22:53:26"
}

GET /v1/orderbook?symbol=ETH/USDT
▲ Bids          ▼ Asks
3,142.89  40.45    3,151.16  40.45
3,136.60  33.69    3,157.46  33.69
3,130.32  22.72    3,163.76  22.72
— Spread: 8.27 —
```

SQLite Ledger — Double-Entry Bookkeeping

Code	Name	Type	Balance
1000	Cash - USD	asset	\$100,000.00
1100	BTC Wallet	asset	0.50 BTC
4000	Revenue - Trading	revenue	\$1,245.30
5000	Fees Paid	expense	\$33.72

✓ Double-entry balanced | Trades logged: 1

AST Static Code Audit

- CRITICAL L5 Missing try-except on urllib.request.urlopen()
- WARNING L11 O(n²) nested loop: for i in range(len(prices))
- INFO L4 Missing type annotation for parameter 'url'
- CRITICAL L3 Use of eval() detected – security risk

Complexity Score: 14 | 4 critical, 1 warning, 2 info



Getting Started

```
# Download the script
python arbitrage_academy.py

# Output:
#  [✓] Web Dashboard: http://localhost:8081
#
#  ┌────────────────────────────────────────────────────────────────────────────────┐
#  │                                PRINCIPAL STRATEGIST                                │
#  └────────────────────────────────────────────────────────────────────────────────┘
#
#  WELCOME, TRAINEE.
#  Type "start" to begin Day 1.
```

CLI Commands

COMMAND	ACTION
<code>start</code>	Begin current day (launches exchange + workspace)
<code>eod</code>	Trigger End-of-Day evaluation
<code>advance</code>	Move to next day (requires score 100)
<code>run</code>	Execute your solution file
<code>audit</code>	Run AST code audit on your solution
<code>ledger</code>	View account balances and trades
<code>exchange</code>	Check exchange server status
<code>status</code>	Show training progress summary
<code>reset</code>	Reset state back to Day 1
<code>help</code>	Show command help



Requirements

- **Python 3.8 or later** (3.11+ recommended for best f-string support)
- **No pip packages** — uses `cmd`, `http.server`, `sqlite3`, `ast`, `threading`, `json`, `urllib`, `subprocess`, and other stdlib modules
- **Port 8080** (Mock Exchange) and **8081** (Web Dashboard) must be free



Licensing

TIER	PRICE	AUDIENCE
Individual	\$79	Single developer, personal use
Professional	\$249	Freelancers, commercial use
Academy	\$750	Classrooms, bootcamps (up to 30 seats)
Enterprise	\$2,500+	Unlimited seats, custom modules, white-label

See [EULA.md](#) for full terms.

? FAQ

Q: Do I need to install anything? A: No. Python 3.8+ ships with everything required.

Q: Is this a course or a tool? A: Both. It's a simulator that generates 90 days of progressive training exercises, evaluates your code, and won't let you advance unless you score 100.

Q: Can I modify the curriculum? A: Days 1–3 are hand-crafted. Days 4–90 use procedural generation. You can modify the generator or add your own curricula.

Q: Does it work on Windows/Mac/Linux? A: Yes. Tested on Python 3.8–3.13 on Windows, macOS, and Linux.

Q: What if I find a bug? A: Individual license holders get 30 days of email support. Higher tiers include SLA-backed support.



About NovaCap Financial Technologies

NovaCap Financial Technologies Ltd. is a simulated FinTech firm used for training environments. All trading data is synthetic. All standards referenced (CMA, SOCPA, IFRS) are real professional accounting frameworks used to structure realistic compliance scenarios.

Built with Python standard library. No dependencies. No shortcuts.

Generated by NovaCap Principal Strategist — Python Standard Library Only